

Tyler Vander Mooren

Problem 1: The car garage

Table 1: Entity types

Entity type name	Attribute	Primary key? (yes/no)*	Multivalued? (yes/no)*	Derived? (yes/no)*	Composed? (yes/no)*
Car	vin	Yes			
	license_plate				
	make				
	model				
	color				
	year				
Client	client_id	Yes			
	first_name				
	last_name				
	phone_number				
	address				Yes
Address	street_number				
	street_name				
	apt_unit				
	city				
	state				
	zipcode				
Authorized_contact	contact_id	Yes			
	first_name				
	last_name				
	phone_number				
Supplies	supplies_id	Yes			
	description				
	brand				
	cost_price				
	sale_price				
	stock_quantity				
Repair	repair_id	Yes			
	repair_date				
	problem_description				
	total_cost			Yes	
Procedure	procedure_code	Yes			
	description				
	cost				

Table 2: Relationship types

Entity type (left side)	Relationship type name	Entity type (right side)	Cardinality (1:1, 1:N, N:1, or N:N)	Identifying? (yes/no)*
Client	owns	Car	1:N	
Client	authroizes	Authroized_contact	1:N	
Car	has	Repair	1:N	Yes
Repair	includes	Procedure	1:N	
Procedure	uses	Supply	N:N	
Client	buys	Supply	N:N	

*For yes/no cells, blank means no.

Table 3: Relationship Set attributes

Relationship type	Attribute name	Primary key? (yes/no)*	Multivalued? (yes/no)*	Derived? (yes/no)*	Composed? (yes/no)*
Repair uses Supply	quantity_used				
Repair uses Supply	discount_applied				
Client buys Supply	quantity				
Client buys Supply	purchase_date				

*For yes/no cells, blank means no.

Problem 2: The library

Table 1: Entity types

Entity type name	Attribute	Primary key? (yes/no)*	Multivalued? (yes/no)*	Derived? (yes/no)*	Composed? (yes/no)*
User	user_id	Yes			
	first_name				
	last_name				
	age				
	education_level				
Student	user_id	Yes			
	course				
	year_of_entry				
Teacher	user_id	Yes			
	department				
	course_taught				
	degree				
Guest	user_id	Yes			
Publication	isbn	Yes			
	title				
	issue_number				
	year				
	available_copies				

Entity type name	Attribute	Primary key? (yes/no)*	Multivalued? (yes/no)*	Derived? (yes/no)*	Composed? (yes/no)*
Periodical	isbn	Yes			
	organizers				
	subtitle				
	volume				
	number				
Book	isbn	Yes			
	authors		Yes		
	publisher				
	city				
Loan	loan_id	Yes			
	checkout_date				
	expected_return_date				
	acutal_return_date				
	fine_amount			Yes	
Reading	reading_id	Yes			
	checkout_datetime				
	return_datetime				
Request	request_id	Yes			
	request_date				
	expiration_date				

*For yes/no cells, blank means no.

Table 2: Relationship types

Entity type (left side)	Relationship name	Entity type (right side)	Cardinality (1:1, 1:N, N:1, or N:N)	Identifying? (yes/no)*
User	borrows	Book	N:N	Yes
User	reads	Publication	N:N	Yes
Teacher	requests	Book	N:N	Yes
Publication	inherits(is?)	Book/Periodical	1:1	
User	inherits(is?)	Student/Teacher/Guest	1:1	

*For yes/no cells, blank means no.

Table 3: Relationship Set attributes

Relationship type	Attribute name	Primary key? (yes/no)*	Multivalued? (yes/no)*	Derived? (yes/no)*	Composed? (yes/no)*
borrows	loan_id	Yes			
	checkout_date				
	expected_return_date				
	actual_return_date				

Relationship type	Attribute name	Primary key? (yes/no)*	Multivalued? (yes/no)*	Derived? (yes/no)*	Composed? (yes/no)*
	fine_amount			Yes	
reads	reading_id	Yes			
	checkout_date_time				
	return_date_time				
requests	request_id	Yes			
	request_date				
	expiration_date				

**For yes/no cells, blank means no.*